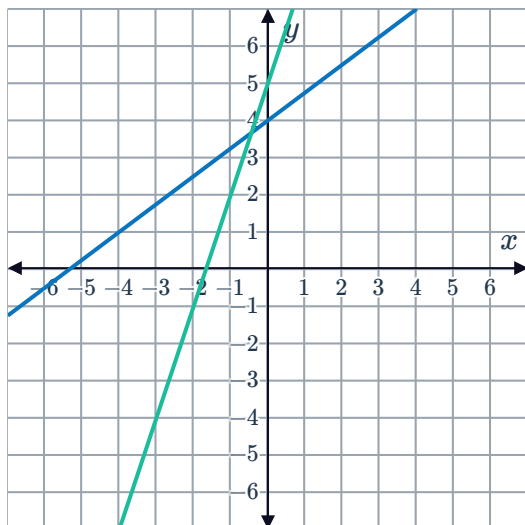


Systems of linear equations

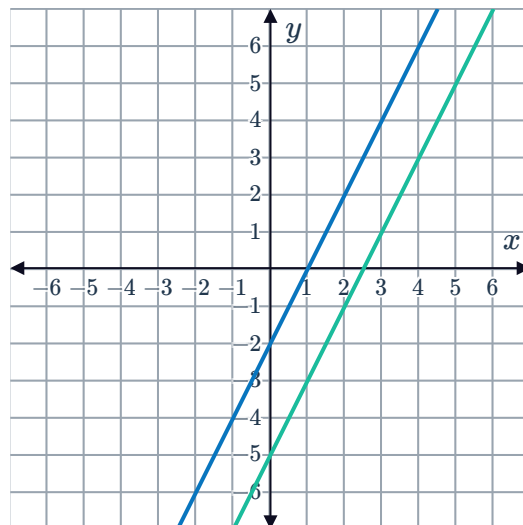


- 1 State the number of solutions for each of the following graphed system of two linear equations:

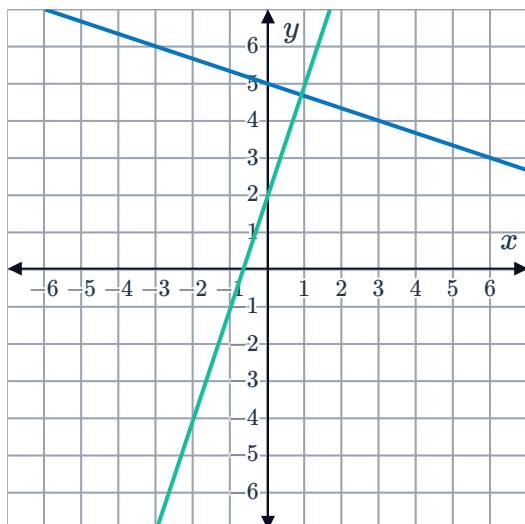
a



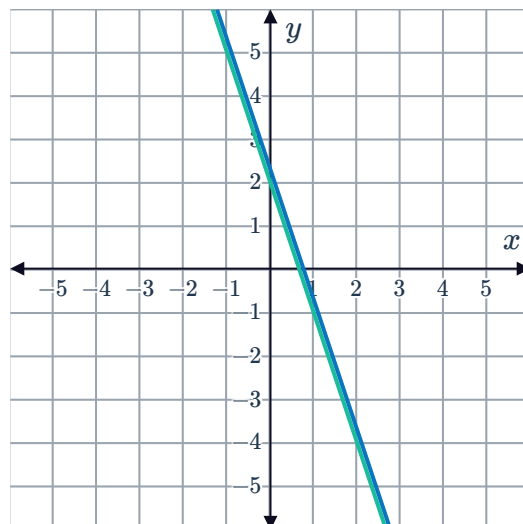
b



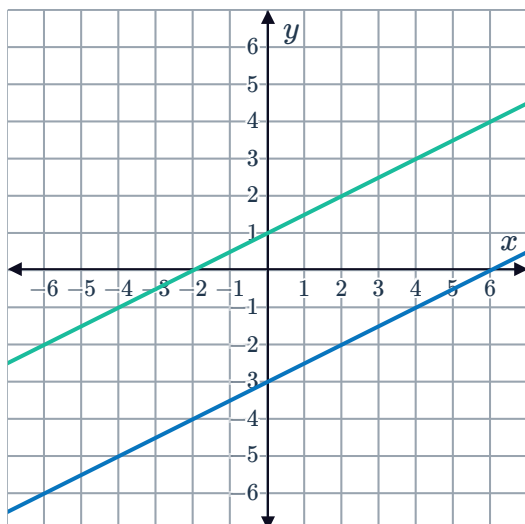
c



d



e

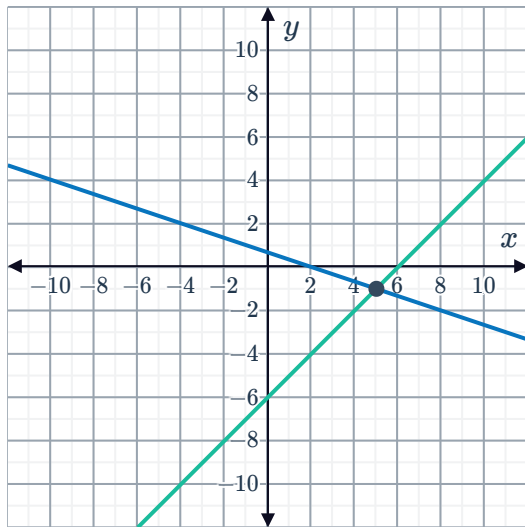


2 Consider the following graphs of systems of linear equations:

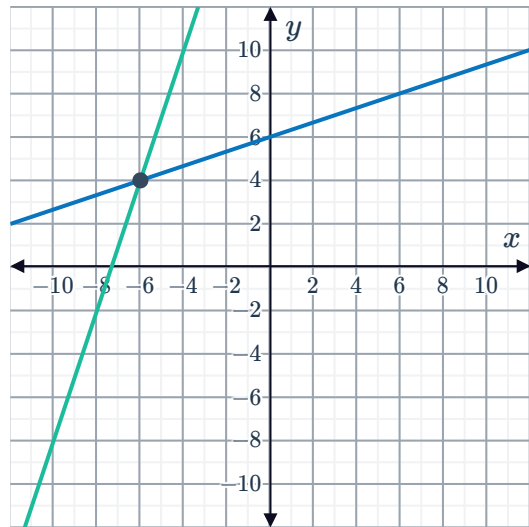


- How many solutions does this system of equations have?
- Write down the solution(s) to the system of equations.

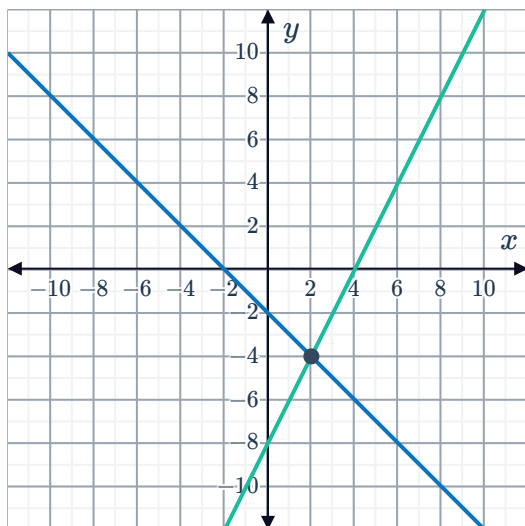
a



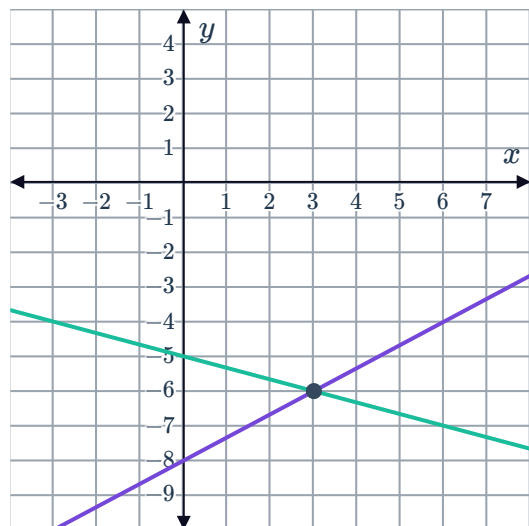
b



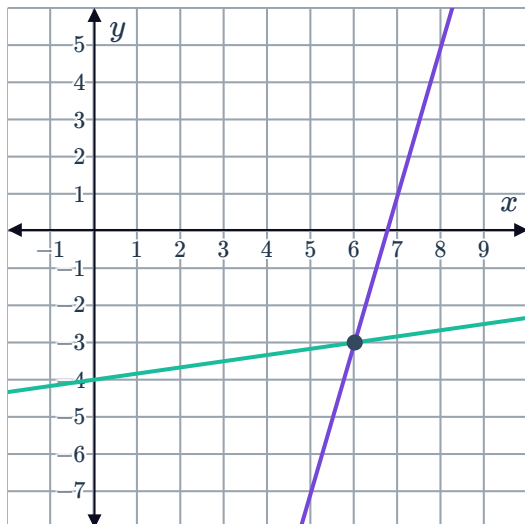
c



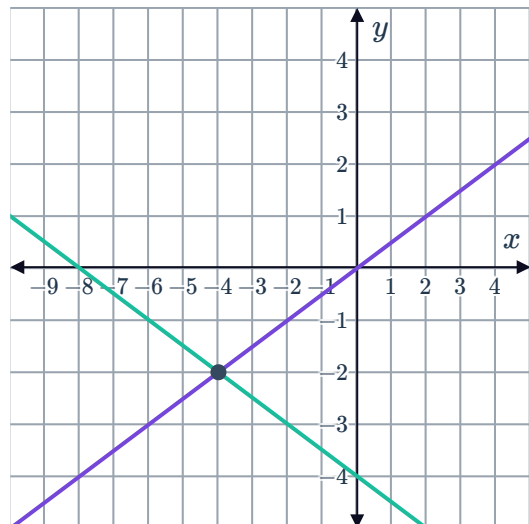
d



e



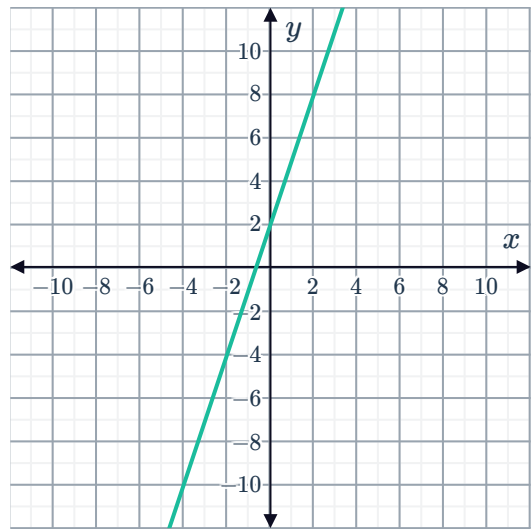
f



- 3 Consider the graph of the equation $y = 3x + 2$:



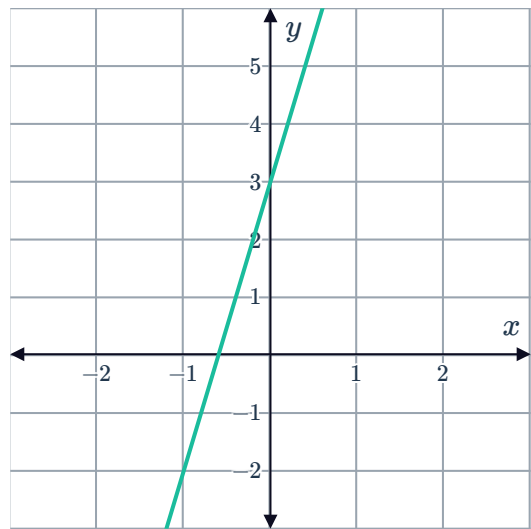
If a second line intersects this line at the point $(0, 2)$, can we determine the equation of the second line? Explain your answer.



- 4 Consider the graph of the equation $y = 5x + 3$:

A second line, $y = mx + b$, intersects this line at the one point $(0, 3)$.

- a State the value of b . Explain your answer.
- b State the value that m cannot be equal to. Explain your answer.



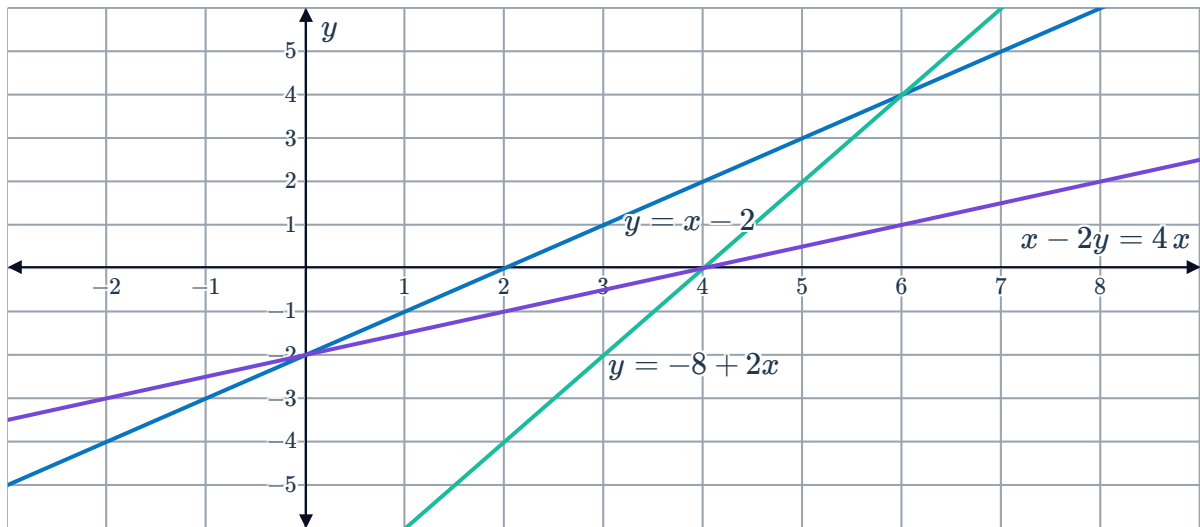
- 5 A system of linear equations has no solutions. One of the equations of the system is $y = -3x - 2$. Determine whether the following could be the other equation of the system.

- a $y = -3x - 3$
- b $y = \frac{x}{3} - 3$
- c $y = -\frac{x}{3} - 2$
- d $y = 3x + 2$



Solve systems graphically

6 Use the given graph to solve each pair of simultaneous equations:



- a** $y = x - 2$
 $y = -8 + 2x$
b $y = x - 2$
 $x - 2y = 4$
c $y = -8 + 2x$
 $x - 2y = 4$

7 For each pair of linear equations:

- i** Sketch the lines on the same number plane.
ii State the coordinates which satisfies both equations.

- a** $L_1 : y = 4x - 4$ **b** $L_1 : y = x + 9$ **c** $L_1 : y = x + 3$ **d** $L_1 : 2x - 4y = -8$
 $L_2 : y = 8 - 2x$ $L_2 : y = -x - 9$ $L_2 : y = -x + 3$ $L_2 : -4x + 2y = -8$
e $L_1 : y = 3x + 6$ **f** $L_1 : y = -x - 3$ **g** $L_1 : y = 4x - 3$
 $L_2 : y = -x + 2$ $L_2 : y = -1$ $L_2 : y = 4 - 3x$

8 Consider the following linear equations:

$$L_1 : y = -2x - 5$$

$$L_2 : y = -2x + 4$$

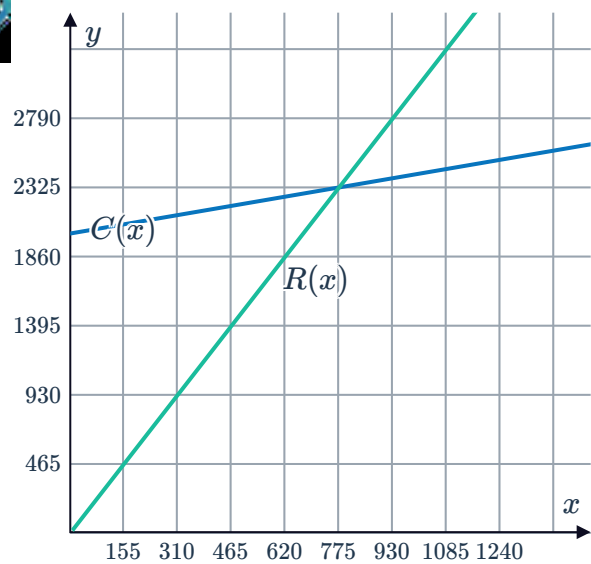
- a** Sketch L_1 and L_2 on the same number plane.
b Is there a coordinate that satisfies the two equations simultaneously? Explain your answer.



Applications

- 9 A rectangular zone is to be 3 m longer than it is wide, with a total perimeter of 18 m.
- Let y represent the length of the rectangle and x represent the width. Construct two equations that represent this information.
 - Sketch the two lines on the same number plane.
 - Hence, find the length and width of the rectangle.
- 10 A band plans to record a demo at a local studio. The cost of renting studio A is \$250 plus \$50 per hour. The cost of renting studio B is \$50 plus \$100 per hour. The cost, y , in dollars of renting the studios for x hours can be modelled by the linear system:
- Studio A: $y = 50x + 250$
 - Studio B: $y = 100x + 50$
- Sketch the two lines on the same number plane.
 - State the coordinate which satisfies both equations.
 - What does the coordinate from part (d) mean?
- 11 Michael plans to start taking an aerobics class. Non-members pay \$4 per class. Members pay a \$10 one-time fee, but only have to pay \$2 per class. The monthly cost, y , of taking x classes can be modelled by the linear system:
- Non-members: $y = 4x$
 - Members: $y = 2x + 10$
- Sketch the two lines on the same number plane.
 - State the coordinate which satisfies both equations.
 - What does the coordinate from part (b) mean?
- 12 The cost of manufacturing toys, C , is related to the number of toys produced, n , by the formula $C = 400 + 2n$. The revenue, R , made from selling n toys is given by $R = 4n$.
- Sketch the graphs of cost and revenue on the same number plane.
 - How many toys need to be produced for the revenue to equal the cost?
 - State the meaning of the y -coordinate of the point of intersection.

- 13 Given the cost function $C(x) = 0.4x + 201$ and the revenue function $R(x) = 3x$, find the coordinates of the point of intersection, or the break-even point.



- 14 The two equations $y = 3x + 35$ and $y = 4x$ represent Laura's living expenses and income from work respectively.
- Find the point of intersection of the two equations.
 - Sketch both equations on the same number plane.
 - State the meaning of the point of intersection of the two lines.
- 15 The two equations $y = 4x + 400$ and $y = 6x$ represent a company's revenue and expenditure respectively.
- Find the point of intersection of the two equations.
 - Sketch both equations on the same number plane.
 - State the meaning of the point of intersection of the two lines.